

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

SHORT ANSWER TEST

Code of Conduct

I ASWATHA V- 192525390 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ASWATHA

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

CO1 AT 3

CLOUD COMPUTING - CSA1520

SHORT ANSWER TEST

NAME : Aswatha . V

REG NO : 192525390

- Q.1. Define cloud computing, explain any 4 characteristics, what makes it different from traditional on-premise.
- Q.2. Evolution of cloud computing from hardware evolution to internet software evolution.
- Q.3. Role of server virtualization & Diff between virtualization as enabling technology & cloud computing as a service model.
- Q.4. Role of data centers in delivering cloud services.
- Q.5. Diff IaaS, PaaS, SaaS, & XaaS with example.

7/10/26
Wed.

CO1 AT3

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1. Cloud Computing :

INTRODUCTION:

It is a modern method of providing computing resources through the Internet. It allows users to access servers, storage, database, software without owning all the physical infrastructure.

CHARACTERISTICS OF CLOUD :

- self-service : users can obtain computing resources whenever required.
- Broad Network access :
Available through internet.
Can access from diff locations.
- Elasticity :
resources can be increased or decreased according to the demand.

TRADITIONAL COMPUTING :

- large initial hardware investment is required
- complex organisation
- high maintainance is required.

But cloud requires less physical infrastructure and it usually follows pay-as-you-use.

CONCLUSION:

Thus cloud provides flexible, scalable & convenient access to computing resources. More efficient & adaptable than traditional on-premise computing.

2. Evolution of cloud computing

INTRODUCTION:

The evolution of cloud computing from 1950's till now has lots of improvements in technologies & internet gradually.

(1) Mainframe Computing:

- Founded by IBM Company
- highly powerful & handles massive data.
- Highly expensive.
- No fault tolerance.

(2) Distributed systems:

Composition of multiple independent system but all are depicted as a single entity

- Effective usage
- Shares resources from one to another system.

The main disadvantage is that all system must be present in same geographical location.

(3) Cluster Computing :

The main host is connected one or more network with high bandwidth.

- Easy addition of nodes
- cheaper than mainframe.

(4) Grid Computing :

- Data is placed in grid structure.
- Different systems were placed at different geographical locations and these all were connected through internet.

(5) Utility Computing :

Defines service provisioning techniques for services such as compute service along with other major services such as storage, infrastructure through pay-per-use basis.

(6) Virtualization

It refers to the process of creating virtual layer over the hardware which allows user to run multiple instances.

(7) Cloud Computing :

- cloud providers combined data centers, network and internet technologies
- users could access computing resources whenever required.
- pay-per-use & resources needed accordingly.

3. Server Virtualization:

- It is the process of creating virtual versions of physical servers using software.
- Each virtual machine can operate independently with its own OS & applications.

ROLE IN CLOUD:

- Better Resource Utilization:
 - ⊙ Multiple VM can be runned on one server.
 - ⊙ CPU & memory are used efficiently
 - ⊙ It reduces unused hardware capacity.
- Cost Reduction:
 - ⊙ Server requirement is less.
 - ⊙ shared infrastructure
- scalability:
 - ⊙ VM can be created, modified or removed
 - ⊙ Can provide additional resources when needed
 - ⊙ flexible cloud services.

VIRTUALIZATION AS ENABLING TECHNOLOGY

- used to divide & manage physical computing
- Helps create VM & efficient use of hardware
- supports cloud infrastructure

CLOUD AS SERVICE MODEL:

→ provides resources & applications to users through internet

Eg: IaaS, PaaS, SaaS [software as a service]

H. Data Centers in Cloud.

Data centers is facility which contains servers, storage, network equipment, power system, etc.

It provides the physical infrastructure required for cloud services.

ROLES:

(1) cloud infrastructure:

→ These data centers works with ^{power & cooling} cloud to store data in servers for users.

(2) Data storage:

→ uses files, databases, applications and backups are used in data centers.

(3) Networking:

→ High speed networks connects servers & storage system.

(4) security:

→ They use firewalls, encryption etc techniques to protect huge data's from theft or lost.

Data storage
Net
Security

Data centers are very important in cloud computing because they provide servers, storage memory space, security, Networks etc.

5. Diff IaaS, PaaS, SaaS, XaaS with examples.

(i) IaaS - Infrastructure as a service

- Provide VM, storage & networking
- Manages operating systems & applications.

Eg. AWS

(ii) PaaS - Platform as a service

- provide platforms for developing applications.
- Manages servers and operating systems

Eg. Google search engines.

(iii) SaaS - Software as a service.

- provide software application through Internet.

Eg: Gmail.

(iv) XaaS - Anything as a service.

- provides different types of technology services through cloud.

→ Includes services beyond IaaS, PaaS & SaaS.

→ Eg. Microsoft - Azure.